



REPUBLIC OF THE PHILIPPINES
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CITY OF DASMARIÑAS

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City College of Dasmariñas



ISMAT3000L: Quantitative Methods Laboratory

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APPENDIX 4

LAB ACTIVITY 4	Names (LN, FNI in alphabetical order): Group #: _____ Date Submitted: _____ Section: <u>BSIS-301</u>	Score:
Date Performed: _____		

Results and Observations

DATASETS USED

- ☐ ExamScores_Normal.csv
- ☐ DailySales_Skewed.csv

Activity 1 - NORMALITY ASSESSMENT Graphical Methods

For each dataset, attach the following:

Dataset	Histogram	Boxplot	Q-Q Plot
Exam Scores	☐	☐	☐
Daily Sales	☐	☐	☐

(Attach on a separate page)



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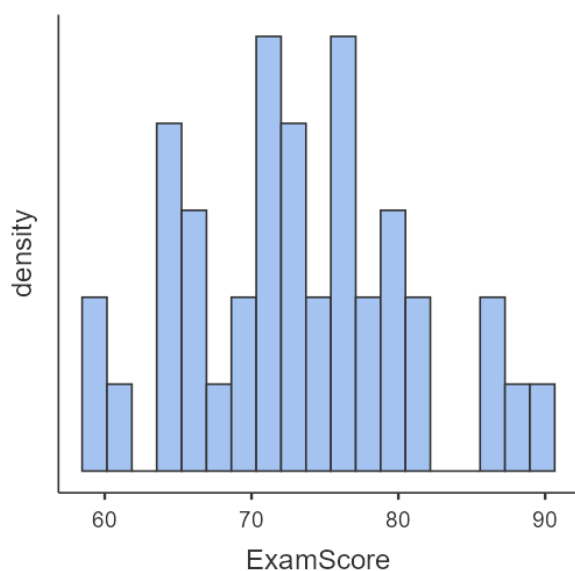


ExamScore

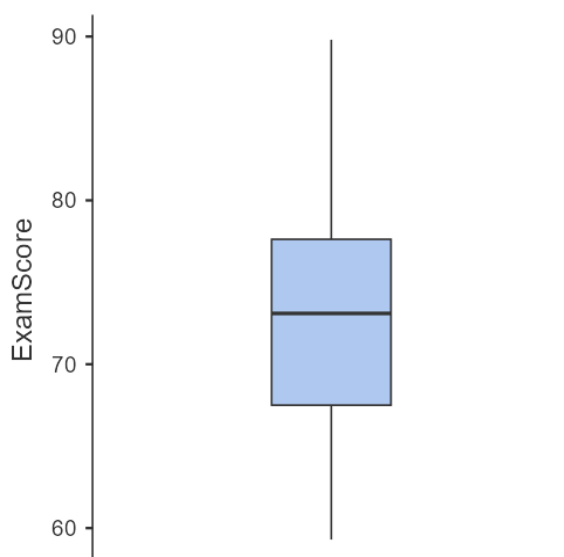
HISTOGRAM:

Plots

ExamScore



BOXPLOT:





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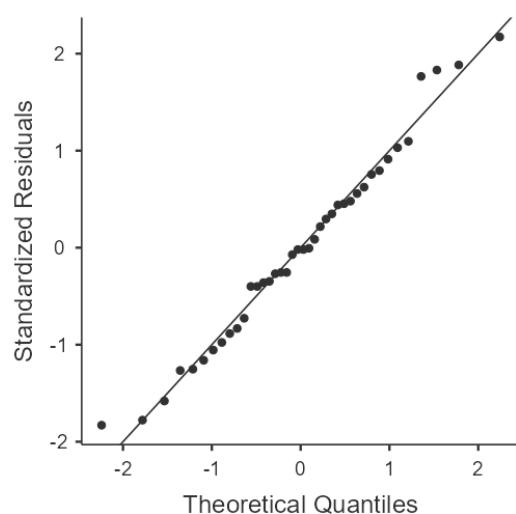
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Q-Q PLOT:

Plots

ExamScore

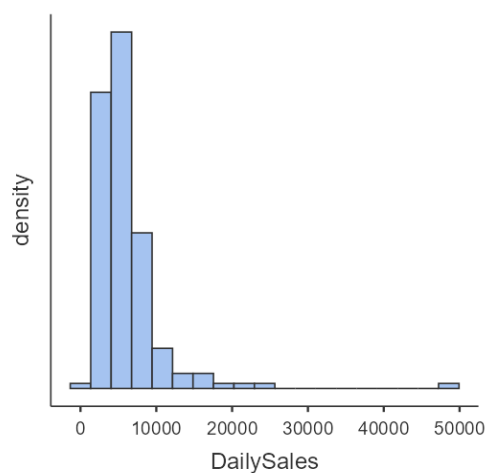


DailySales

HISTOGRAM:

Plots

DailySales



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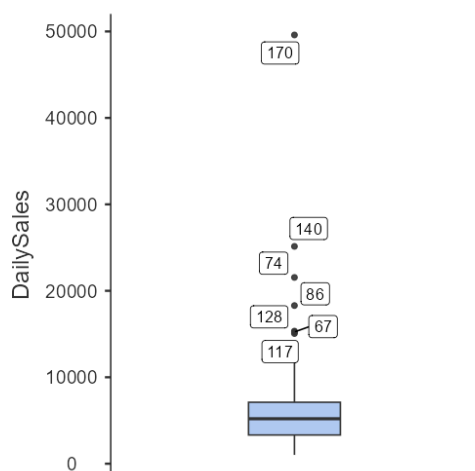


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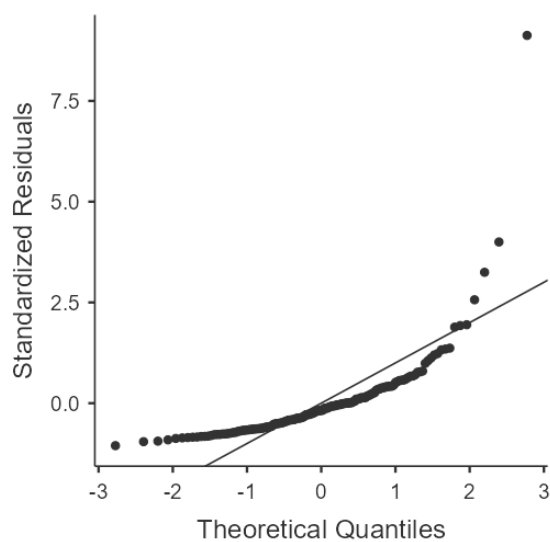
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BOXPLOT:



Q-Q PLOT:



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Activity 2 Normality Tests

Complete the table below.

Dataset	Shapiro–Wilk p	KS / Lilliefors p	Anderson–Darling p	Normal? (Yes/No)
Exam Scores	0.662	0.982	Accept normality	YES
Daily Sales	0.001	0.00004	Reject normality	NO

	Post-Lab Questions
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1. Which dataset appears approximately normal based on the Q-Q plot?

- Exam Scores dataset appears approximately normal

2. Which normality tests rejected normality for Daily Sales?

- Shapiro-Wilk, KS / Lilliefors, and Anderson–Darling all rejected normality.

3. Why do multiple tests give stronger evidence than a single test?

- Different Test detects different deviations, so consistent results increase reliability

4. Why might large samples reject normality even for small deviations?

- Large samples make tests very sensitive, so even minor deviations become statistically significant.

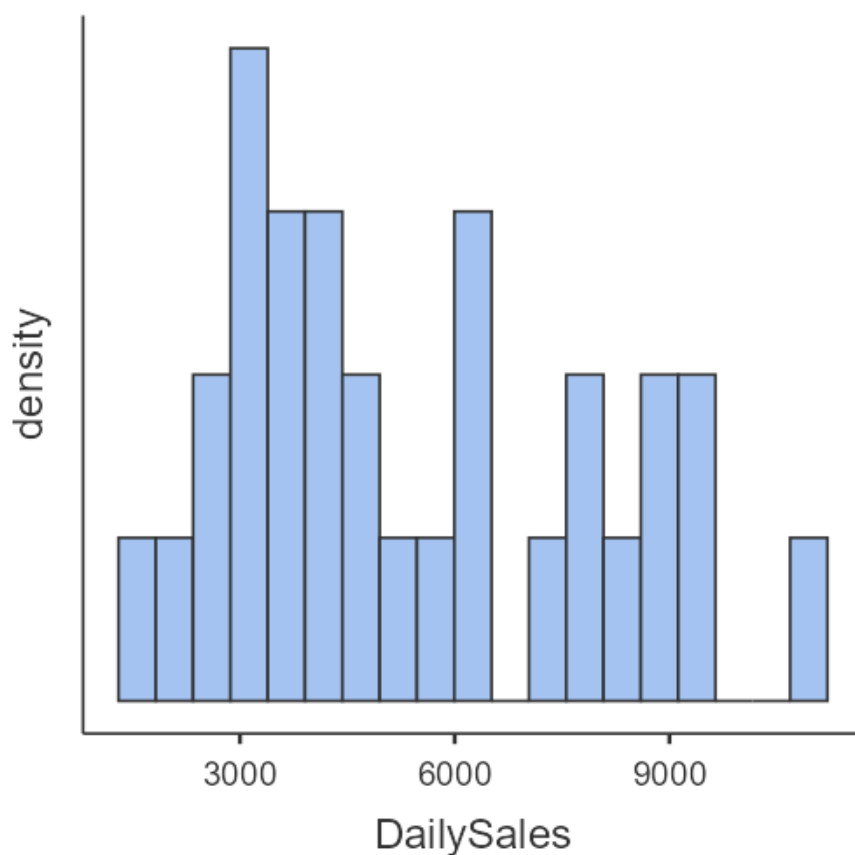




Activity 3 CENTRAL LIMIT THEOREM

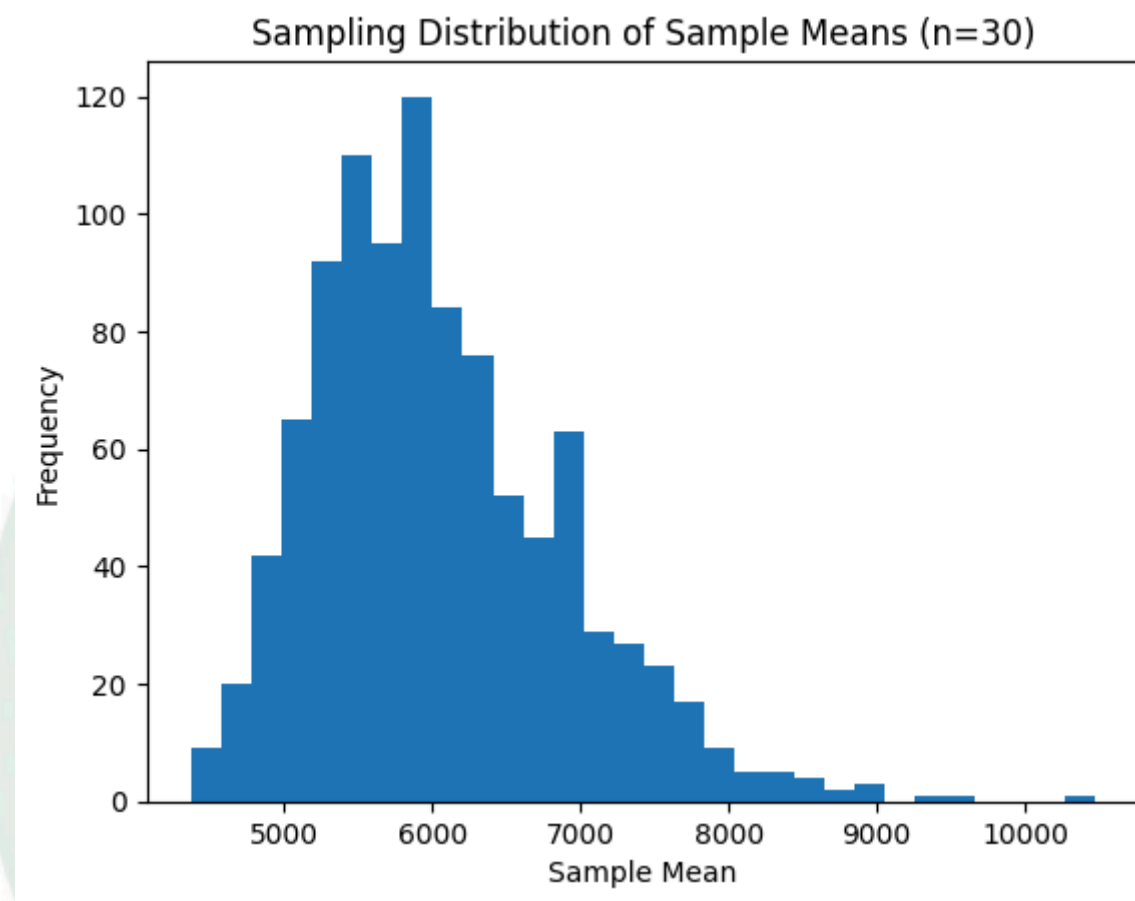
1. Draw repeated samples of size 30 from Daily Sales

DailySales





2. Plot the distribution of sample means



3. Does the sampling distribution appear approximately normal? Why?

Yes, the average of the samples looks almost normal even if the original daily sales data is not. This happens because of the Central Limit Theorem. It says that when the sample size is large (30 or more), the distribution of the sample averages becomes close to normal.



Activity 4 CONFIDENCE INTERVALS

t-Confidence Interval (Education Dataset)

Quantity	Value
Sample Mean	73.2
Sample SD	7.62
n	40
t-critical	2.023
95% CI	70.75, 75.65

z-Confidence Interval (Business Dataset)

Quantity	Value
Sample Mean	6046
Sample SD	4772
n	180
z-critical	1.96
95% CI	5349, 6743



	Post-Lab Questions
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1. Why is the t-distribution used for exam scores but not for daily sales?

- Exam Scores use smaller sample theory with unknown population SD

2. How does the CLT justify using z-confidence intervals for skewed data?

- It states that sampling distribution becomes normal for large sample

3. Which confidence interval is wider and why?

- Daily Sales CI is wider due to higher variability

4. How can confidence intervals help school administrators or managers?

- It can help the school administrators or managers by providing a range of expected values for better decision-making.



	Summary and Conclusions
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Summarize what you learned about estimation, normality testing, CLT, and confidence intervals.

This activity shows how important statistics is when making decisions. We use estimation to guess unknown values of a large group (population) by using smaller data (samples). By checking the data using graphs and tests, we learned that not all real-life data follow a normal (perfect bell-shaped) pattern, and this affects what statistical method we should use.

The Central Limit Theorem explains that even if the original data is not normal, the average of many samples will become close to normal when the sample size is large. Because of this, we can still use methods like z-confidence intervals.

Confidence intervals help show a range where the true value is likely to be found. This helps people make better decisions in areas like education and business.

Overall, the activity helped improve understanding of estimation, checking data patterns, sampling, and how to interpret confidence intervals in real-life situations.



APPENDIX 5

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Date Performed: _____	Group #: _____ Date Submitted: _____ Section: <u>BSIS-301</u>	

Results and Observations

DATASETS USED

ExamScores_Normal.csv (Education context)

DailySales_Normal.csv (Business context)

ACTIVITY 1 — ONE-SAMPLE Z-TEST (BUSINESS CONTEXT)

A retail company claims that the **average daily sales** is **₱50,000**.

1. Hypotheses

State the hypotheses clearly.

- H_0 : (Null): $\mu = 50,000$
- H_1 : (Alternative): $\mu \neq 50,000$

Type of test: ☐ Two-tailed ☒ Left-tailed ☐ Right-tailed

Significance level (α): 0.05



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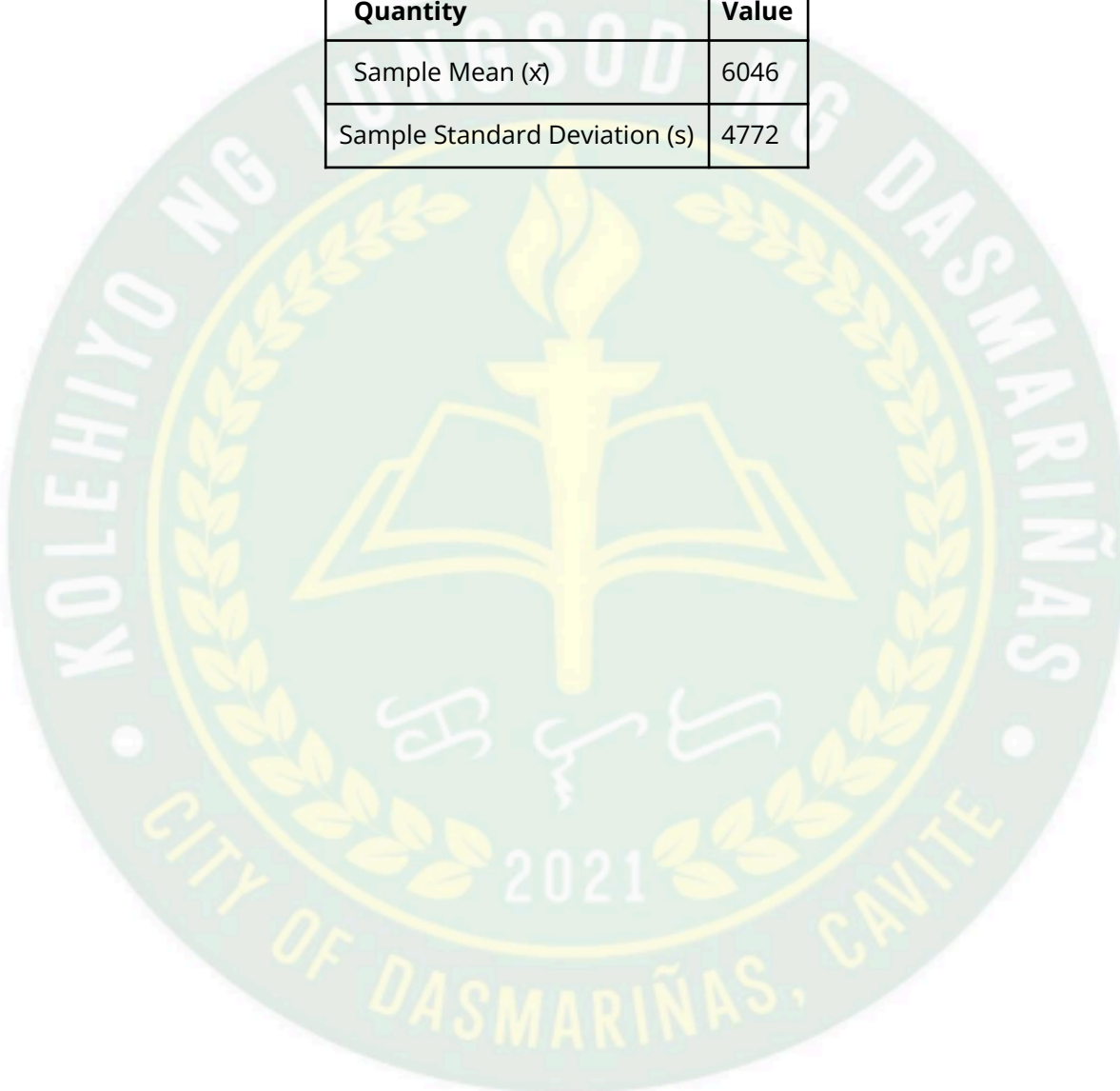
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2. Sample Information

Complete the table using your dataset.

Quantity	Value
Sample Mean ($\bar{x}$)	6046
Sample Standard Deviation (s)	4772



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